Effect of Background Music Type and Time Awake on Typing Performance

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Background

Studies have shown that both music and time since waking affect cognitive and motor performance [1, 2]. Calm music, such as Mozart, may improve concentration [1], while high-tempo music could be overstimulating. Similarly, cognitive performance varies based on time of day [2]. Therefore, typing performance offers a measurable way to evaluate these effects. This study aims to experimentally assess the combined influence of music type and time awake on typing speed and accuracy.

Objective & Hypothesis

To test whether typing performance depends on both music tempo and time awake.

Null hypothesis ($\mathbf{H_0}$): No main or interaction effects of music or time awake on typing performance.

Alternative hypothesis ($\mathbf{H_1}$):

(a) Classical music ($\sim 60-80$ BPM) improves typing speed and accuracy relative to silence and hardstyle (~ 150 BPM).
(b) Performance peaks after 4–8 hours of wakefulness.
(c) Favorable music mitigates performance decline during late-evening conditions.

Variables

Factors (predictors):

- **Music Type:** Silence (control), Classical music ($\sim$ 60–80 BPM), Hardstyle music ($\sim$ 150 BPM)
- **Time Awake:** Within 1 hour of waking, $4-8$ hours awake (midday), $\geq$ 12 hours awake

Response Variables:

- **Typing speed (WPM):** Calculated as keystrokes divided by five.
- **Typing accuracy (%):** Correct keystrokes / (correct keystrokes + corrections), accounting for both errors and backspaces.

Experimental Design

A 3×3 full factorial design with repeated measures. Each participant will complete all nine combinations of music and time-awake conditions in randomized order. The study will run over a one-week period, allowing flexibility in scheduling and giving room for occasional missed trials. A minimum of three participants will be collected; however, a sample size of $\geq 12-15$ ought to be collected to give the study a higher validity. The data will be tracked per participant to distinguish different baselines in typing speed and accuracy. Typing tests (1 minute each) will be conducted using a version of the 10FastFingers test featuring the 200 most common English words to standardize difficulty across sessions. The test will be accessed via a custom-built experiment website, which will also embed music delivery through YouTube, Spotify or local music files. A link to the website will be included in the final report.

Feasibility

A self-hosted website will guide participants by randomizing conditions, embedding music and typing test links, automating typing data, and offering a form to log "time awake". If full automation proves unfeasible, a fallback system using pre-randomized trial sheets and Google Forms will be implemented. Each session will only take a couple of minutes, and participants can complete sessions flexibly over the course of the week.

Gantt Chart

The Gantt Chart in Figure 1 summarizes the planned activities, timeline and tasks for the experiment. The same chart is also available as an online Excel Document at the following URL.

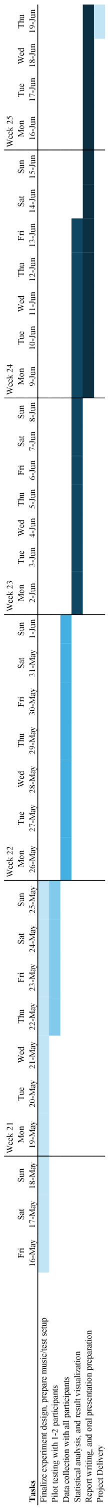

Figure 1: Gantt chart showing experiment timeline and tasks.

References

[1] Leslie A. Angel, Donald J. Polzella and Greg C. Elvers. 'Background music and cognitive performance'. eng. In: *Perceptual and Motor Skills* 110.3 Pt 2 (June 2010), pp. 1059–1064. ISSN: 0031-5125. DOI: 10.2466/pms.110.C.1059-1064.

[2] Shiyang Xu, Miriam Akioma and Zhen Yuan. 'Relationship between circadian rhythm and brain cognitive functions'. In: *Frontiers of Optoelectronics* 14.3 (Apr. 2021), pp. 278–287. ISSN: 2095-2759. DOI: 10.1007/s12200-021-1090-y. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9743892/ (visited on 15th May 2025).